

JESS Thermodynamic Database v8.9

Chemical species in reactions with Polonium

24-Jan-23

Charge	CAS	Count	Molecular formula	Mol. mass
2ThenTriFAcone-1				
-1	---	101	C(8)H(4)F(3)O(2)S(1)	221.174
Acetic-1 Acetate ion				
-1	71-50-1	558	C(2)H(3)O(2)	59.0446
Ag+1 Silver(I) ion				
1	14701-21-4	1534	Ag(1)	107.870
Ag(s) Silver; Silver, cubic				
0	7440-22-4	182	Ag(1)	107.870
Ag2PoO3(s)				
0	---	2	Ag(2)O(3)Po(1)	472.738
Br2(l) Bromine, liquid				
0	---	285	Br(2)	159.808
C(s) Carbon; Graphite; Carbon, hexagonal				
0	7440-44-0	530	C(1)	12.0110
Cl-1 Chloride ion				
-1	16887-00-6	2242	Cl(1)	35.4530
Cl2(g) Chlorine gas				
0	7782-50-5	589	Cl(2)	70.9060

ClO4-1 Perchlorate ion				
-1	14797-73-0	114	Cl(1)O(4)	99.4506
CN-1 Cyanide ion				
-1	57-12-5	658	C(1)N(1)	26.0177
e-1 Electron				
-1	---	3399	E(1)	0.000000
H+1 Hydrogen ion; Proton				
1	12408-02-5	31679	H(1)	1.00794
H+1_Citric-3				
-2	---	93	C(6)H(6)O(7)	190.109
H+1_EDTA-4				
-3	---	70	C(10)H(13)N(2)O(8)	289.222
H+1_NO3-1				
0	---	32	H(1)N(1)O(3)	63.0128
H+1_NTA-3				
-2	---	33	C(6)H(7)N(1)O(6)	189.125
H+1_Po-2 Hydrogen polonide				
-1	---	3	H(1)Po(1)	210.008
H+1_PoO3-2				
-1	---	5	H(1)O(3)Po(1)	258.006
H+1_S-2 Hydrogensulfide ion				
-1	---	140	H(1)S(1)	33.0679

H+1_SO4-2 Hydrogensulfate ion				
-1	---	83	H(1)O(4)S(1)	97.0655
H+1(2)_Po-2 Dihydrogen polonide				
0	---	3	H(2)Po(1)	211.016
H+1(2)_Po-2_(g)				
0	---	1	H(2)Po(1)	211.016
H+1(2)_PoO3-2				
0	---	6	H(2)O(3)Po(1)	259.014
H+1(2)_S-2 Hydrogen sulphide; Hydrogen sulfide				
0	---	58	H(2)S(1)	34.0759
H+1(2)_SeO4-2 Selenic acid				
0	---	6	H(2)O(4)Se(1)	144.977
H+1(2)_SO4-2 Sulphuric acid; Sulfuric acid				
0	---	15	H(2)O(4)S(1)	98.0735
H2(g) Hydrogen gas; Hydrogen				
0	1333-74-0	1564	H(2)	2.01588
H2O Water				
0	7732-18-5	5947	H(2)O(1)	18.0153
Hg+2 Mercury(II) ion; Mercuric ion				
2	14302-87-5	1203	Hg(1)	200.592
Hg(1) Mercury; Quicksilver				

0	7439-97-6	160	Hg (1)	200.592
HgPo (s)				
0	---	2	Hg (1) Po (1)	409.592
I-1 Iodide ion				
-1	20461-54-5	894	I (1)	126.900
I2 (s) Iodine; Iodine, orthorhombic				
0	7553-56-2	244	I (2)	253.800
N2 (g) Nitrogen gas; Nitrogen				
0	7727-37-9	566	N (2)	28.0134
Ni+2 Nickel(II) ion				
2	14701-22-5	5108	Ni (1)	58.6930
Ni (s) Nickel; Nickel, fcc				
0	7440-02-0	148	Ni (1)	58.6930
NiPo (s)				
0	---	2	Ni (1) Po (1)	267.693
NO3-1 Nitrate ion				
-1	14797-55-8	573	N (1) O (3)	62.0049
O2 (g) Oxygen gas; Oxygen				
0	7782-44-7	2559	O (2)	31.9988
OH-1 Hydroxide ion				
-1	14280-30-9	6558	H (1) O (1)	17.0073

Oxalic-2 Oxalate ion				
-2	338-70-5	1241	C(2)O(4)	88.0196
Pb+2 Lead(II) ion				
2	14280-50-3	2437	Pb(1)	207.200
Pb(s) Lead; Lead, cubic				
0	7439-92-1	127	Pb(1)	207.200
PbPo(s)				
0	---	2	Pb(1)Po(1)	416.200
Po-2 Polonide ion				
-2	---	11	Po(1)	209.000
Po+2 Polonium(II) ion				
2	---	31	Po(1)	209.000
Po+2_Br-1(2)_(s) Polonium(II) bromide				
0	---	1	Br(2)Po(1)	368.808
Po+2_Cl-1				
1	---	2	Cl(1)Po(1)	244.453
Po+2_Cl-1(2)				
0	---	2	Cl(2)Po(1)	279.906
Po+2_Cl-1(2)_(s) Polonium(II) chloride				
0	---	2	Cl(2)Po(1)	279.906
Po+2_Cl-1(3)				
-1	---	2	Cl(3)Po(1)	315.359

Po+2_Cl-1 (4)				
-2	---	4	Cl (4) Po (1)	350.812
Po+2_H+1				
3	---	1	H (1) Po (1)	210.008
Po+2_H+1 (2)				
4	---	1	H (2) Po (1)	211.016
Po+2_I-1 (2) (s) Polonium(II) iodide				
0	---	1	I (2) Po (1)	462.800
Po+2_S-2 (s)				
0	---	4	Po (1) S (1)	241.060
Po+2_SO4-2				
0	---	3	O (4) Po (1) S (1)	305.058
Po+2_SO4-2 (s) Polonium(II) sulfate				
0	---	2	O (4) Po (1) S (1)	305.058
Po+4 Polonium(IV) ion				
4	---	76	Po (1)	209.000
Po+4_Acetic-1				
3	---	1	C (2) H (3) O (2) Po (1)	268.045
Po+4_Acetic-1 (2)				
2	---	1	C (4) H (6) O (4) Po (1)	327.089
Po+4_Acetic-1 (3)				
1	---	1	C (6) H (9) O (6) Po (1)	386.134
Po+4_Br-1 (4) (s) Polonium(IV) bromide				

0	---	1	Br (4) Po (1)	528.616
Po+4 _Cl-1				
3	---	1	Cl (1) Po (1)	244.453
Po+4 _Cl-1 (2)				
2	---	1	Cl (2) Po (1)	279.906
Po+4 _Cl-1 (3)				
1	---	1	Cl (3) Po (1)	315.359
Po+4 _Cl-1 (4)				
0	---	1	Cl (4) Po (1)	350.812
Po+4 _Cl-1 (4) _ (s) Polonium(IV) chloride				
0	---	3	Cl (4) Po (1)	350.812
Po+4 _Cl-1 (4) _OH-1				
-1	---	2	Cl (4) H (1) O (1) Po (1)	367.819
Po+4 _Cl-1 (5)				
-1	---	2	Cl (5) Po (1)	386.265
Po+4 _Cl-1 (6)				
-2	---	8	Cl (6) Po (1)	421.718
Po+4 _ClO4-1				
3	---	1	Cl (1) O (4) Po (1)	308.451
Po+4 _ClO4-1 (2)				
2	---	1	Cl (2) O (8) Po (1)	407.901
Po+4 _ClO4-1 (3)				
1	---	1	Cl (3) O (12) Po (1)	507.352
Po+4 _ClO4-1 (4)				

0	---	1	Cl (4) O (16) Po (1)	606.802
Po+4 _CN-1 (6)				
-2	---	4	C (6) N (6) Po (1)	365.106
Po+4 _H+1 (2) _OH-1 (2) _Citric-3 (2)				
-2	---	1	C (12) H (14) O (16) Po (1)	623.234
Po+4 _H+1 (2) _OH-1 (2) _EDTA-4 (2)				
-4	---	1	C (20) H (28) N (4) O (18) Po (1)	821.458
Po+4 _H+1 (2) _OH-1 (2) _NTA-3 (2)				
-2	---	1	C (12) H (16) N (2) O (14) Po (1)	621.264
Po+4 _I-1 (4) _ (s) Polonium(IV) iodide				
0	---	3	I (4) Po (1)	716.600
Po+4 _I-1 (5)				
-1	---	2	I (5) Po (1)	843.500
Po+4 _I-1 (6)				
-2	---	2	I (6) Po (1)	970.400
Po+4 _NO3-1				
3	---	1	N (1) O (3) Po (1)	271.005
Po+4 _NO3-1 _OH-1 (2)				
1	---	1	H (2) N (1) O (5) Po (1)	305.020
Po+4 _NO3-1 (2)				
2	---	1	N (2) O (6) Po (1)	333.010
Po+4 _NO3-1 (2) _OH-1 (2)				
0	---	1	H (2) N (2) O (8) Po (1)	367.025
Po+4 _NO3-1 (3)				

1	---	1	N (3) O (9) Po (1)	395.015
Po+4 _NO3-1 (3) _OH-1 (2)				
-1	---	1	H (2) N (3) O (11) Po (1)	429.029
Po+4 _NTA-3 (2)				
-2	---	1	C (12) H (12) N (2) O (12) Po (1)	585.234
Po+4 _OH-1				
3	---	3	H (1) O (1) Po (1)	226.007
Po+4 _OH-1 (2)				
2	---	12	H (2) O (2) Po (1)	243.015
Po+4 _OH-1 (2) _Oxalic-2 (2)				
-2	---	1	C (4) H (2) O (10) Po (1)	419.054
Po+4 _OH-1 (2) _Tartaric-2 (2)				
-2	---	1	C (8) H (10) O (14) Po (1)	539.159
Po+4 _OH-1 (3)				
1	---	4	H (3) O (3) Po (1)	260.022
Po+4 _OH-1 (4)				
0	---	3	H (4) O (4) Po (1)	277.029
Po+4 _OH-1 (4) _ (s)				
0	---	2	H (4) O (4) Po (1)	277.029
Po+4 _OH-1 (6)				
-2	---	2	H (6) O (6) Po (1)	311.044
Po+4 _Oxalic-2				
2	---	2	C (2) O (4) Po (1)	297.020
Po+4 _Oxalic-2 (2)				
0	---	2	C (4) O (8) Po (1)	385.039

Po+4 _SO4-2				
2	---	3	O (4) Po (1) S (1)	305.058
Po+4 _SO4-2 (2)				
0	---	3	O (8) Po (1) S (2)	401.115
Po+4 _SO4-2 (2) _H2O_ (s)				
0	---	3	H (2) O (9) Po (1) S (2)	419.131
Po+4 _SO4-2 (3)				
-2	---	1	O (12) Po (1) S (3)	497.173
Po+6 _OH-1 (2)				
4	---	1	H (2) O (2) Po (1)	243.015
Po (aq) Polonium, elemental, aquated				
0	---	1	Po (1)	209.000
Po (g) Polonium gas				
0	---	1	Po (1)	209.000
Po (s) Polonium; Polonium, primitive cubic				
0	7440-08-6	66	Po (1)	209.000
Po2 (g)				
0	---	1	Po (2)	418.000
PoO . PoO . (NO3) 3 . OH (s) Polonium(IV)hydroxynitrate				
0	---	3	H (1) N (3) O (12) Po (2)	653.021
PoO . PoO . OSeO4 (s)				
0	---	4	O (7) Po (2) Se (1)	608.959

PoO.PoO.OSO4 (s) Polonium(V) sulfate				
0	---	3	O(7)Po(2)S(1)	562.056
PoO+2				
2	---	55	O(1)Po(1)	224.999
PoO+2_2ThenTriFAcne-1				
1	---	1	C(8)H(4)F(3)O(3)Po(1)S(1)	446.173
PoO+2_2ThenTriFAcne-1(2)				
0	---	1	C(16)H(8)F(6)O(5)Po(1)S(2)	667.347
PoO+2_Cl-1(2)_OH-1				
-1	---	3	Cl(2)H(1)O(2)Po(1)	312.913
PoO+2_Cl-1(4)				
-2	---	4	Cl(4)O(1)Po(1)	366.811
PoO+2_CN-1(2)_(s)				
0	---	3	C(2)N(2)O(1)Po(1)	277.035
PoO+2_H+1_EDTA-4				
-1	---	1	C(10)H(13)N(2)O(9)Po(1)	514.221
PoO+2_H+1(2)_Citric-3(2)				
-2	---	1	C(12)H(12)O(15)Po(1)	605.218
PoO+2_H+1(2)_NTA-3(2)				
-2	---	1	C(12)H(14)N(2)O(13)Po(1)	603.249
PoO+2_NO3-1				
1	---	2	N(1)O(4)Po(1)	287.004
PoO+2_NO3-1(2)				
0	---	2	N(2)O(7)Po(1)	349.009

PoO+2_NO3-1 (3)				
-1	---	3	N (3) O (10) Po (1)	411.014
PoO+2_OH-1				
1	---	4	H (1) O (2) Po (1)	242.007
PoO+2_OH-1 (2)				
0	---	4	H (2) O (3) Po (1)	259.014
PoO+2_OH-1 (2)_(s) Polonium dioxide, hydrated				
0	---	2	H (2) O (3) Po (1)	259.014
PoO+2_Oxalic-2 (2)				
-2	---	1	C (4) O (9) Po (1)	401.039
PoO+2_SeO4-2				
0	---	3	O (5) Po (1) Se (1)	367.960
PoO+2_SeO4-2 (2)				
-2	---	3	O (9) Po (1) Se (2)	510.921
PoO+2_SO4-2				
0	---	3	O (5) Po (1) S (1)	321.057
PoO+2_SO4-2 (2)				
-2	---	3	O (9) Po (1) S (2)	417.115
PoO+2_SO4-2 (3)				
-4	---	3	O (13) Po (1) S (3)	513.172
PoO+2_Tartaric-2 (2)				
-2	---	1	C (8) H (8) O (13) Po (1)	521.144
PoO2 (s) Polonium(IV) oxide; Polonium dioxide				
0	7446-06-2	15	O (2) Po (1)	240.999

PoO3-2				
-2	---	18	O(3)Po(1)	256.998
PoO3(s) Polonium trioxide				
0	---	6	O(3)Po(1)	256.998
S-2 Sulfide ion; Sulphide ion				
-2	14127-58-3	466	S(1)	32.0600
S(s) Sulphur, alpha; Sulphur, orthorhombic; Sulphur; Sulfur, alpha; Sulfur, orthorhombic; Sulfur; Sulfur-Rhmb				
0	7704-34-9	619	S(1)	32.0600
Se(s) Selenium, trigonal; Selenium, grey; Selenium, black; Selenium, gamma; Selenium, hexagonal (sic)				
0	7782-49-2	140	Se(1)	78.9630
SeO4-2 Selenate ion				
-2	---	106	O(4)Se(1)	142.961
SO4-2 Sulfate ion; Sulphate ion				
-2	14808-79-8	1270	O(4)S(1)	96.0576
Tartaric-2 Tartrate ion				
-2	---	393	C(4)H(4)O(6)	148.072
Zn+2 Zinc(II) ion				
2	23713-49-7	4977	Zn(1)	65.3820
Zn(s) Zinc; Zinc, hexagonal				
0	7440-66-6	131	Zn(1)	65.3820

ZnPo (s)				
0	---	2	Po (1) Zn (1)	274.382